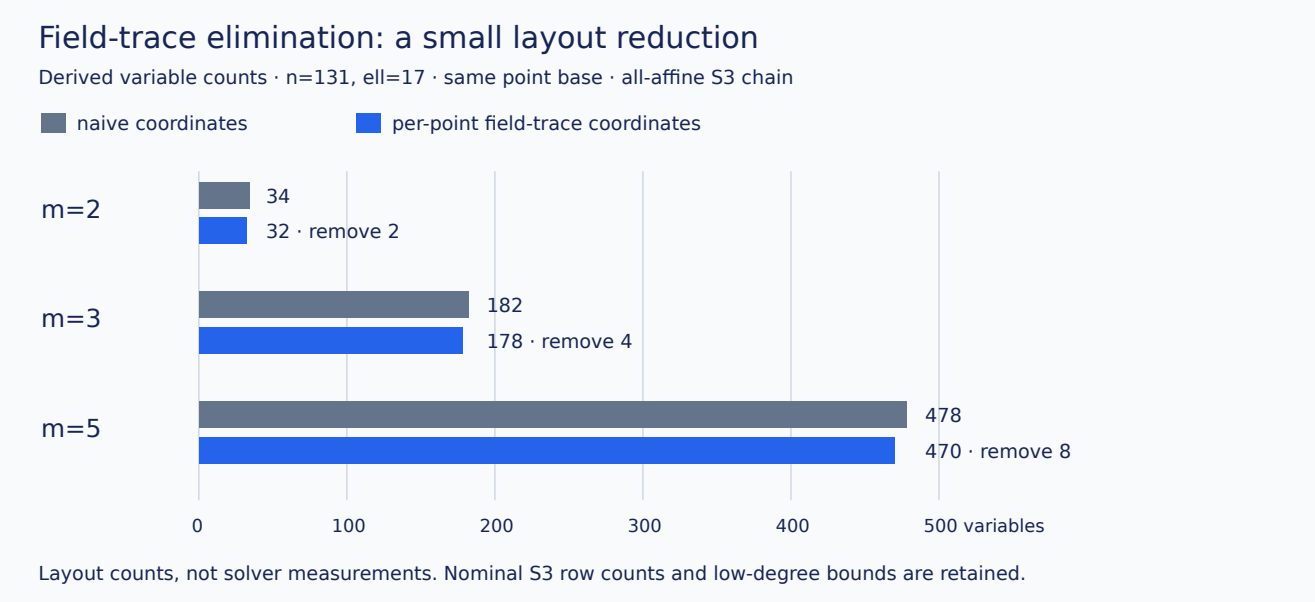
Trace-zero relation systems beyond Frobenius orbits

2026-11-09 · Isogeny-labs/crypto · bounded structural check · unsigned local evidence

Result: A small per-coordinate field-trace elimination is available. The full trace-zero Weil-restriction construction does not by itself establish a smaller matched relation-finding system for ECC2K-130. Twenty-two local unit tests pass; no solver benchmark or signed verification is asserted.



Keep the trace maps distinct

For E_0: $y^2+xy=x^3+1$ and $\tau(x,y)=(x^2,y^2)$, group trace S_n sums n Frobenius conjugate points and takes values in $E(F_2)$. The trace-zero carrier in Res(E) has dimension $n-1$. At $n=131$, $E(F_2)$ has four points and the trace is onto, so its kernel has order $r=680564733841876926932320129493409985129$. This is already the challenge subgroup.

The different bit Tr(x) characterizes 2E, not 4E. Exact toy checks at n=3,5,7,9 give group-trace kernel sizes 1,11,29,127 and field-x-trace kernel sizes 2,22,58,254. These membership conditions are not interchangeable.

In $Z[\tau]$, $\tau^2+\tau+2=0$, and $\tau-1=\bar{\tau}^2$ has norm four. Gorla–Massierer's exact sequence maps $E(F_{2^n})$ onto T_n using $\tau-1$. Since $[4]=\tau^2(\tau-1)$, this image is precisely cofactor clearing. The audit checks the ring identity, independent point-count recurrence, pinned degree-83/131 orders and toy image/kernel equality.

Matched coordinates

Keep $B=\{P \text{ in } T_n: x(P) \text{ in } V\}$, the target and arity fixed. Let $\ell=\dim(V)$, $\varepsilon=\operatorname{rank}(\mathrm{Tr} \text{ restricted to } V)$. An all-affine S3 chain has $m\ell+(m-2)n$ variables. Field-trace coordinates for summands and affine intermediates give $m(\ell-\varepsilon)+(m-2)(n-1)$. This preserves legitimate affine-chain relations. Infinity intermediates still need their own patterns.

In the challenge basis $z^{131}+z^{13}+z^2+z+1$, field trace is supported at positions 0 and 129. For V spanned by $1,z,\ldots,z^{16}$, $\varepsilon=1$. The figure gives derived layout counts. Nominal component rows remain $(m-1)n$ and degree bounds remain quadratic for m=2, cubic for a chain. These are not instantiated Groebner systems or measured solving degrees.

If V already lies in the field-trace kernel, $\varepsilon=0$: no summand bit disappears. Choosing a larger or different base is not a matched comparison.

The full trace-zero formulation

Gorla–Massierer section 3 uses two coordinates per point, n−1 summands, and 2n−1 equations in 2(n−1) variables. At q=2,n=131 the literal factor base contains at most four abscissae/eight affine points. Its 130-term sums occupy at most C(137,7)=153802846692 points. A uniform target in the pinned r-point group has coverage below 2^{-90}. This only concerns that fixed base and arity, not adaptive, larger, nonlinear or orbit-expanded constructions.

At matched m=130 with two coordinates per summand, naive direct S_{131} already has 260 coordinate variables. GM also has 260, with 261 equations including 130 trace-membership equations. A chained S3 representation has 17028 coordinate/intermediate variables and 16899 nominal components. That change removes chain auxiliaries and trades degree; it is not a trace-zero dimension gain.

Formal unreduced S_{131} degree is 2^{129} per argument. After reduction over F_2, Boolean degree is bounded by the variable count. Do not turn a formal degree bound into measured complexity. The paper fixes n in its q-asymptotic advantage and reports exponential growth in n. Its practical cases concern degrees 3 and 5.

Scope and reproduction

Native command: cargo test –locked –manifest-path research/trace_zero_relation_check_20261109/native/Cargo.toml – –nocapture –test-threads=1. The small crate reuses production field arithmetic without rebuilding the root library. Four mechanism checks plus 18 field tests pass locally. checks.txt and SOURCES.sha256 retain output and dependencies.

No timing, relation yield, actual solving degree, full-DLP result or signed independent verdict is established. A stronger claim needs an explicit low-degree trace-zero chart on a matched base, exact polynomial construction, lifting/infinity checks and a controlled solver comparison. Existing checkpoint/EKS access blocks that separate benchmark. Other nonlinear Z[tau] methods remain open.

Primary sources: Gorla–Massierer, Index Calculus in the Trace Zero Variety, sections 2–3 their compression paper, section 3 Galbraith–Granger–Merz–Petit, SAC 2020, introduction and section 5 Kosters–Yeo, Proposition 4.2. README.md gives repository pointers and all comparison assumptions.

Knowledge-base tools and wrapper executable were unavailable; source fallback is recorded in the session log. No novelty or universal impossibility claim.